



Green Future Advisor Agent

AI-Powered Sustainable Living Guidance Assistant

1M1B AI for Sustainability Virtual Internship

(In Collaboration with IBM SkillsBuild & AICTE)

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Project Focus:

Using AI to promote sustainable lifestyle choices and climate-conscious decisions.

Project Overview

 **Green Future Advisor Agent** is an AI-powered sustainability assistant designed to help individuals make environmentally responsible decisions in their daily lives.

The agent provides personalized guidance on:

- Sustainable lifestyle choices
- Carbon footprint awareness
- Energy and resource conservation
- Waste reduction practices
- Climate-friendly actions

Goal:

To use AI as a decision-support tool that helps users understand sustainability and adopt practical eco-friendly habits.



Problem Statement

Many people want to adopt sustainable lifestyles but lack personalized guidance and awareness about how their daily choices impact the environment.

Current challenges:

- Limited understanding of carbon footprint and its reduction
- Difficulty finding practical sustainable solutions
- Lack of personalized recommendations based on individual needs
- Low awareness about climate-friendly habits

Problem Statement:

How might we use AI to provide personalized sustainability guidance so that individuals can make more informed and environmentally responsible choices?



SDG Alignment

Primary SDG:

SDG 13: Climate Action

Supports climate awareness and encourages actions that help reduce environmental impact.



Secondary SDGs:

SDG 12: Responsible Consumption and Production

Promotes sustainable consumption habits, waste reduction, and responsible resource usage.

Secondary SDGs:

SDG 11: Sustainable Cities and Communities

Encourages sustainable living practices that contribute to healthier and more sustainable communities.



Target Users & Impact Area

Target Users

The Green Future Advisor Agent is designed for:

- Students and young individuals interested in sustainable living.
- Individuals looking to reduce their environmental impact.
- Communities seeking awareness about eco-friendly practices.

Impact Area

- Improves sustainability awareness
- Helps users make climate-conscious decisions
- Encourages reduction of carbon footprint
- Promotes responsible use of resources and sustainable habits



AI Solution Overview

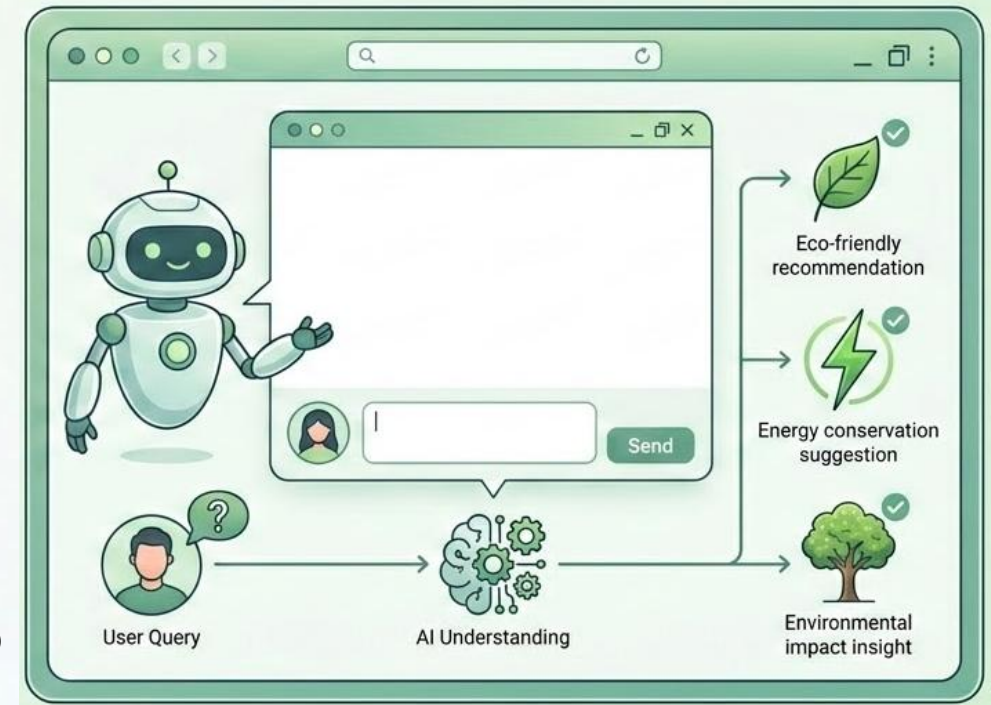
🌱 **Green Future Advisor Agent** is an AI-powered conversational assistant that provides personalized sustainability recommendations based on user needs and queries.

Solution Features:

- Understands user sustainability-related questions
- Provides practical eco-friendly recommendations
- Guides users on reducing environmental impact
- Supports sustainable lifestyle decision-making
- Converts sustainability knowledge into simple actionable steps

AI Approach:

An agent-based AI workflow that uses conversational intelligence to deliver personalized sustainability guidance.



How AI is Used in the Project

AI Implementation

The Green Future Advisor Agent uses AI technologies to understand user queries, process sustainability-related information, and generate personalized eco-friendly recommendations.

AI Components Used:

➤ **IBM watsonx Orchestrate**

Used to build and manage the AI agent workflow and handle user interactions.

➤ **IBM watsonx.ai**

Used to support AI model execution and intelligent response generation.

➤ **IBM Granite Models**

Used for language understanding, reasoning, and creating sustainability-focused responses.

➤ **Agentic AI Workflow**

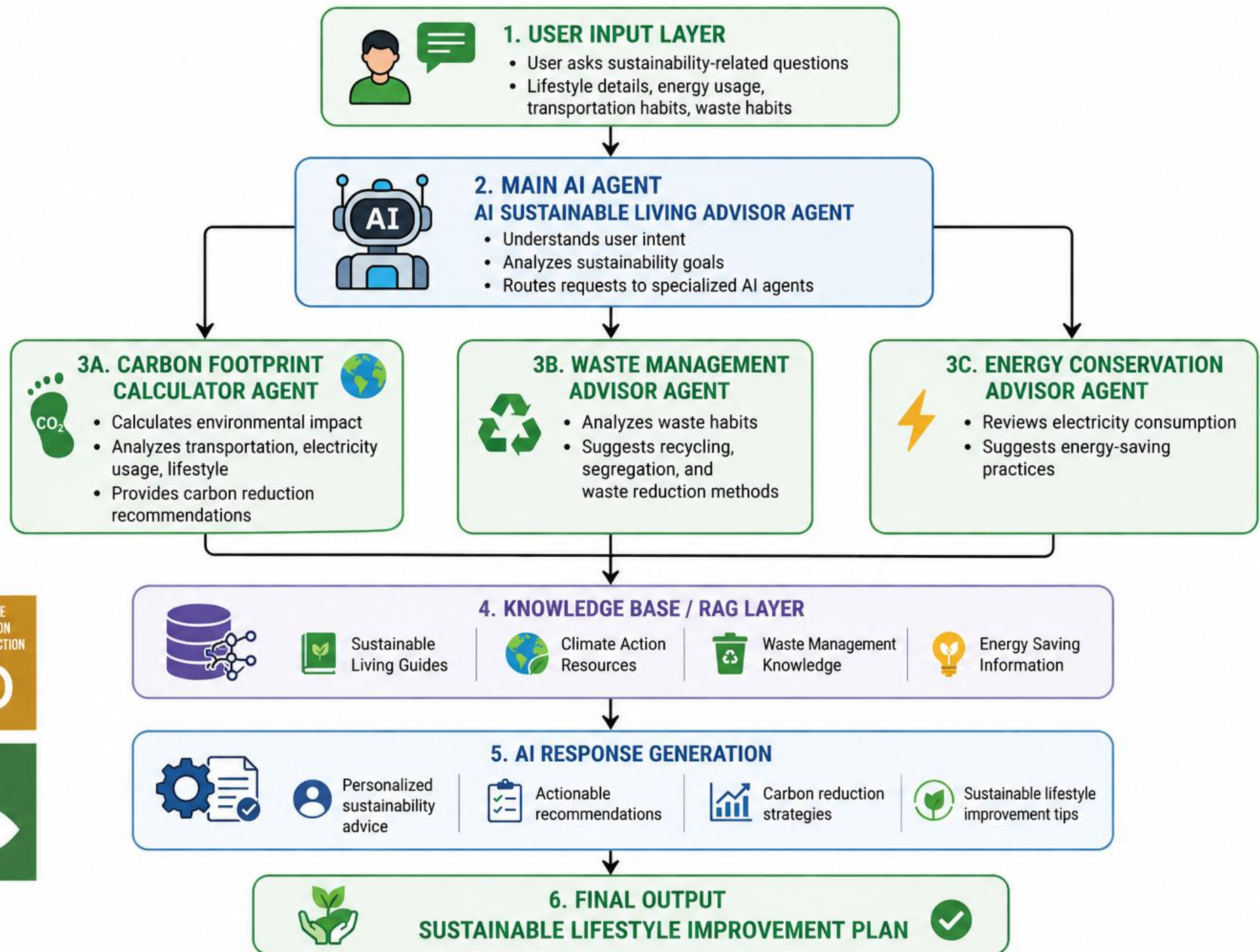
Used to connect user input, AI processing, and recommendation generation.

➤ **Prompt Engineering**

Used to define agent instructions and improve the quality of AI responses.

➤ **Conversational AI**

Used to enable natural interaction between users and the sustainability advisor.



Features of Green Future Advisor Agent

Key Features

Sustainability Guidance

Provides recommendations for adopting eco-friendly lifestyle practices.

Carbon Footprint Awareness

Helps users understand activities that contribute to environmental impact.

Energy Conservation Suggestions

Provides tips for reducing energy consumption and improving efficiency.

Waste Reduction Guidance

Suggests responsible waste management and recycling practices.

Personalized Recommendations

Generates practical solutions based on user queries and daily habits.

AI-Based Interaction

Allows users to communicate naturally with the sustainability assistant.

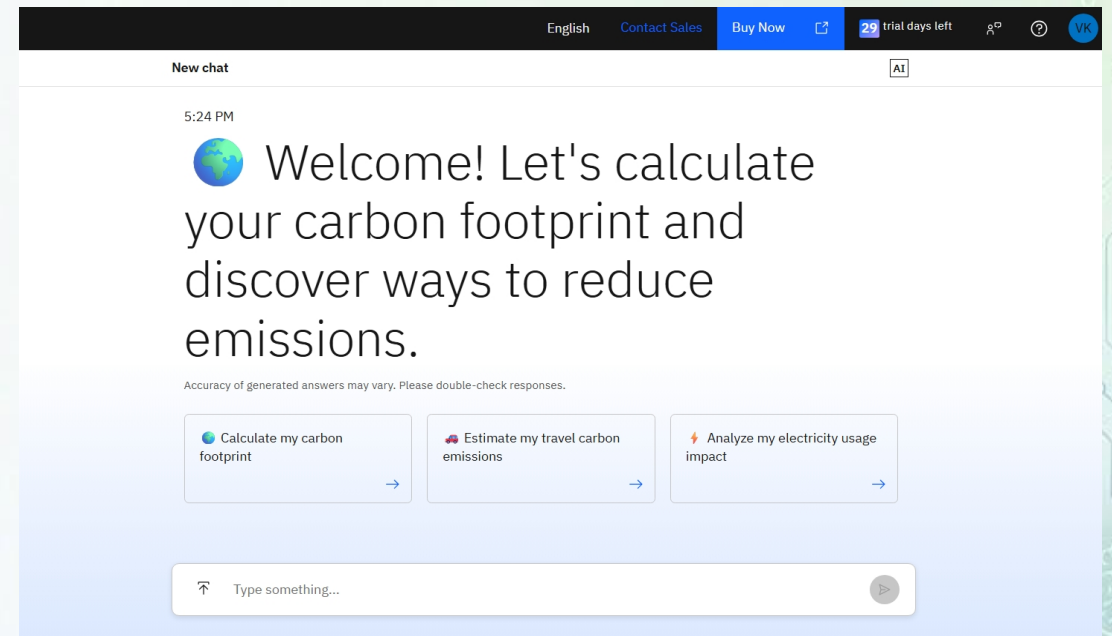
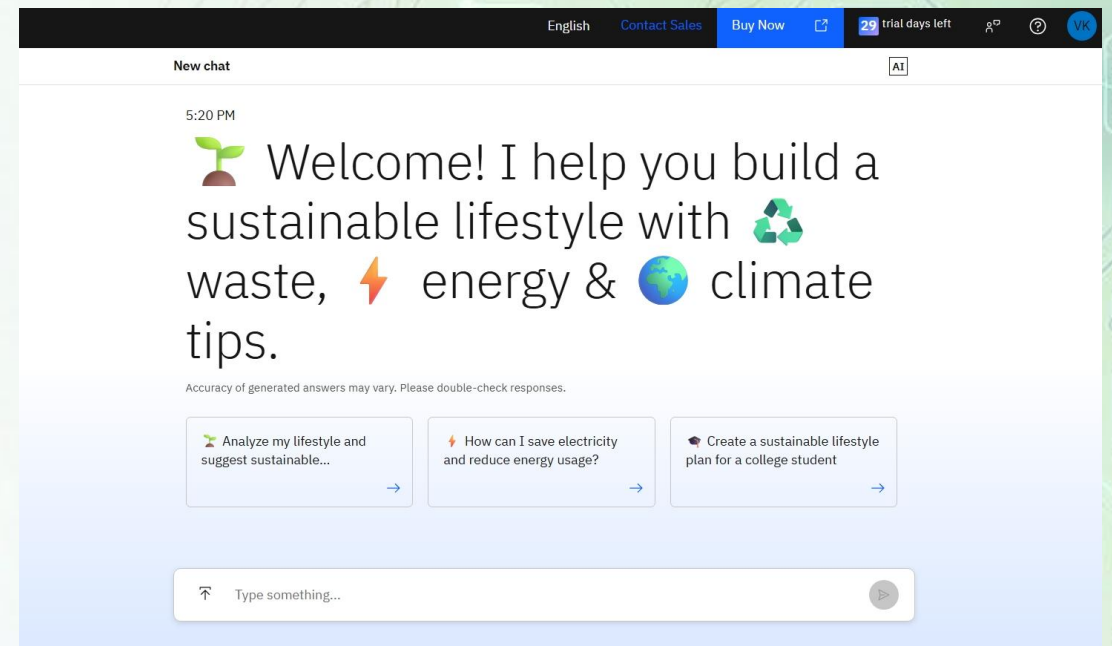
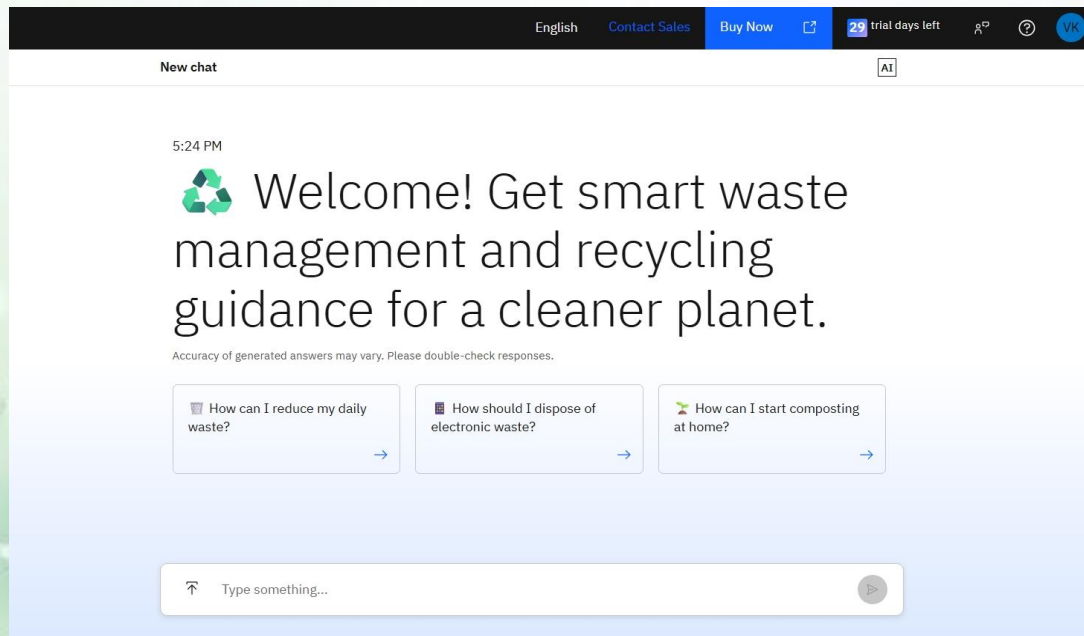


Prototype Demonstration

The project prototype is implemented as an AI sustainability advisor agent.

Prototype Includes:

- AI agent interface for user interaction
- Sustainability-related user queries
- AI-generated personalized recommendations
- Practical action suggestions for sustainable living





Agents



Green Future Advis... x v

Chats



Today

My old laptop and mobile phone are ...

I live in a hostel and generate food wa...

[Create new agent](#)[Manage agents](#)[Agent memory](#) [Preview](#)

My old laptop and mobile phone are not working. Should I throw them away?

AI

You 5:22 PM

My old laptop and mobile phone are not working. Should I throw them away?

Green Future Advisor Agent 5:22 PM | [Show Reasoning](#) v

What to do with your old laptop and mobile phone

Throwing them in the regular trash can release harmful chemicals into soil and water, and you'll miss a chance to get useful material back into the circular economy. Here's a simple, eco-friendly path you can follow.

✓ Action Steps

1. Check if they can be repaired

- Look for loose screws, broken screens, or dead batteries.
- If a simple fix (e.g., replacing a battery or cleaning contacts) could bring them back to life, try that first.

2. Back up and erase your data

- Transfer any important files to a cloud drive or external hard drive.
- Perform a factory reset on the phone and wipe the laptop's hard drive (you can use free tools like *DBAN* for a secure erase).



Type something...





Agents



Green Future Advis... x v

Chats



Today

I use many plastic bottles and bags e...

My old laptop and mobile phone are n...

I live in a hostel and generate food wa...

[Create new agent](#)[Manage agents](#)[Agent memory](#) [Preview](#)

I use many plastic bottles and bags every week. How can I reduce this?

AI

You 5:23 PM

I use many plastic bottles and bags every week. How can I reduce this?



Green Future Advisor Agent 5:23 PM | Show Reasoning v

Let's cut down on those plastic bottles and bags!

Quick wins you can start today

Action	How to do it	Why it helps
Use a reusable water bottle	- Keep a stainless-steel or BPA-free bottle at home, work, and in your bag. - Fill it with tap water or filtered water instead of buying single-use bottles.	- Saves dozens of bottles each month. - Cuts the energy and oil needed to make new plastic.
Carry reusable shopping bags	- Keep 2-3 foldable tote bags in your car, purse, or near the front door. - Choose bags made from recycled or organic cotton, jute, or sturdy polyester.	- Eliminates the need for new plastic grocery bags. - Reduces litter that ends up in oceans.
Refill instead of repurchase	- Look for water refill stations, bulk-dispensing beverage taps, or "fill-your-own" soda stations. - Refill your bottle at home with filtered water.	Keeps the bottle in use longer, decreasing overall plastic production.
	When buying drinks, choose larger reusable containers (e.g., 1-L bottle) and keep a	



Type something...





Green Future Advisor Agent



Model: GPT-OSS 120B — Ope...

[Build with AI](#)[Talk to agent](#)

Running in debug mode

Profile

Knowledge

Toolset

Behavior

Channels [Preview](#)

Welcome message

Customize the welcome message from the agent on the home screen.

Welcome message

95/100

Welcome! I help you build a sustainable lifestyle with  waste,  energy&  climate tips.

Maximum 100 characters allowed in welcome message edit field.

Quick start prompts

Pre-set messages sent to the agent to start the conversation.

[Add prompt](#)

Analyze my lifestyle and suggest sustainable improvements



How can I save electricity and reduce energy usage?



Create a sustainable lifestyle plan for a college student



Agent style

You 5:20 PM



How can I save electricity and reduce energy usage?

 Green Future Advisor Agent 5:20 PM | [Show Reasoning](#) 

Quick wins to save electricity at home



Why it matters

- **Lower carbon emissions** → reduces your personal climate impact.
- **Save money** → less electricity = lower bills.
- **Long-term comfort** → a well-insulated, efficient home stays cooler in summer and warmer in winter without cranking the thermostat.



Action steps you can start to

[Save as test](#)

Type something...



Responsible AI Considerations

The Green Future Advisor Agent follows responsible AI principles to ensure safe, fair, and reliable sustainability guidance.

Fairness

- Provides unbiased sustainability recommendations for different users.
- Avoids assumptions based on personal background.

Transparency

- Gives clear explanations behind recommendations.
- Helps users understand the reasoning behind suggested actions.

Privacy

- Avoids collecting unnecessary personal information.
- Focuses only on user-provided sustainability queries.

Ethics

- Provides responsible and environmentally beneficial guidance.
- Avoids misleading or harmful recommendations.



Expected Impact

The Green Future Advisor Agent aims to create positive environmental impact by helping users make informed and sustainable lifestyle decisions.

Impact Areas:

Environmental Impact

- Encourages reduction of carbon footprint
- Promotes responsible use of resources
- Supports climate-conscious actions

Social Impact

- Increases sustainability awareness among users
- Makes sustainability knowledge easier to access
- Encourages eco-friendly behaviour

Practical Impact

- Provides personalized guidance for daily sustainable choices
- Helps users adopt simple actions with long-term benefits

Overall Impact:

AI-powered guidance can help individuals contribute towards a more sustainable future



Future Scope

The Green Future Advisor Agent can be further improved with advanced features to increase its sustainability impact.

Possible Enhancements:

- **Mobile Application Integration**

Make the sustainability assistant easily accessible through a mobile platform.

- **Personal Carbon Tracking**

Add features to track and analyze individual carbon footprint over time.

- **Multilingual Support**

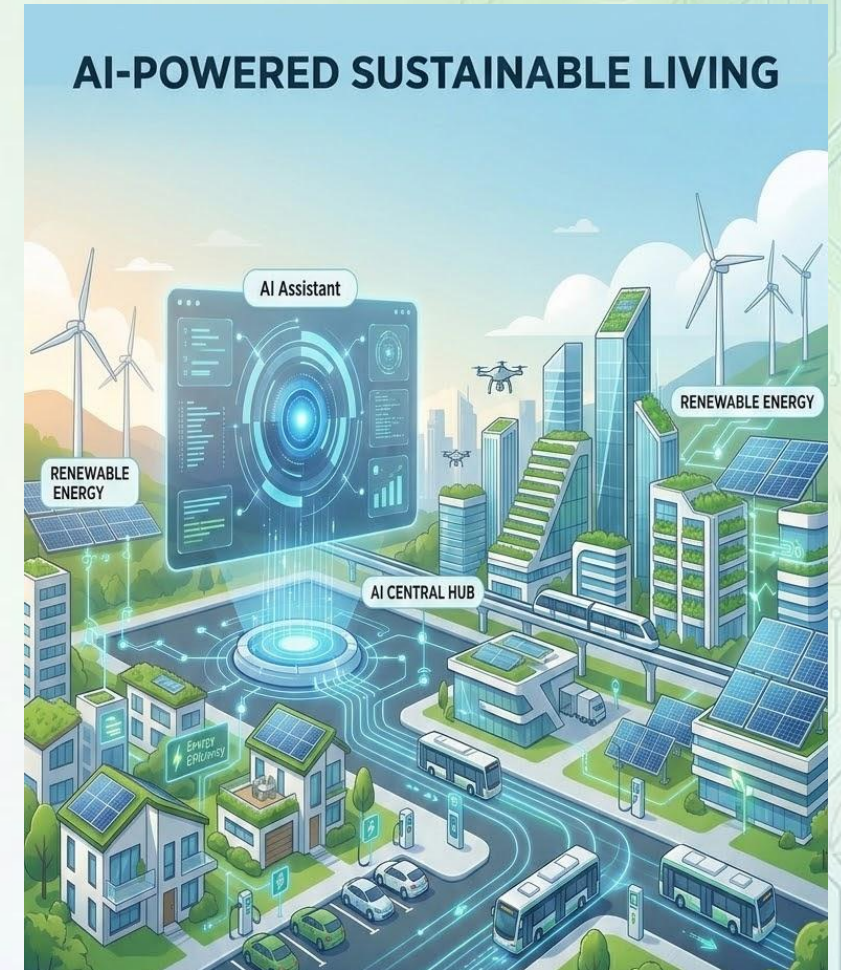
Provide sustainability guidance in multiple languages for wider accessibility.

- **Smart Device Integration**

Connect with smart devices to provide better energy and resource usage insights.

- **Expanded Sustainability Knowledge Base**

Improve recommendations using more sustainability resources and data.



Conclusion

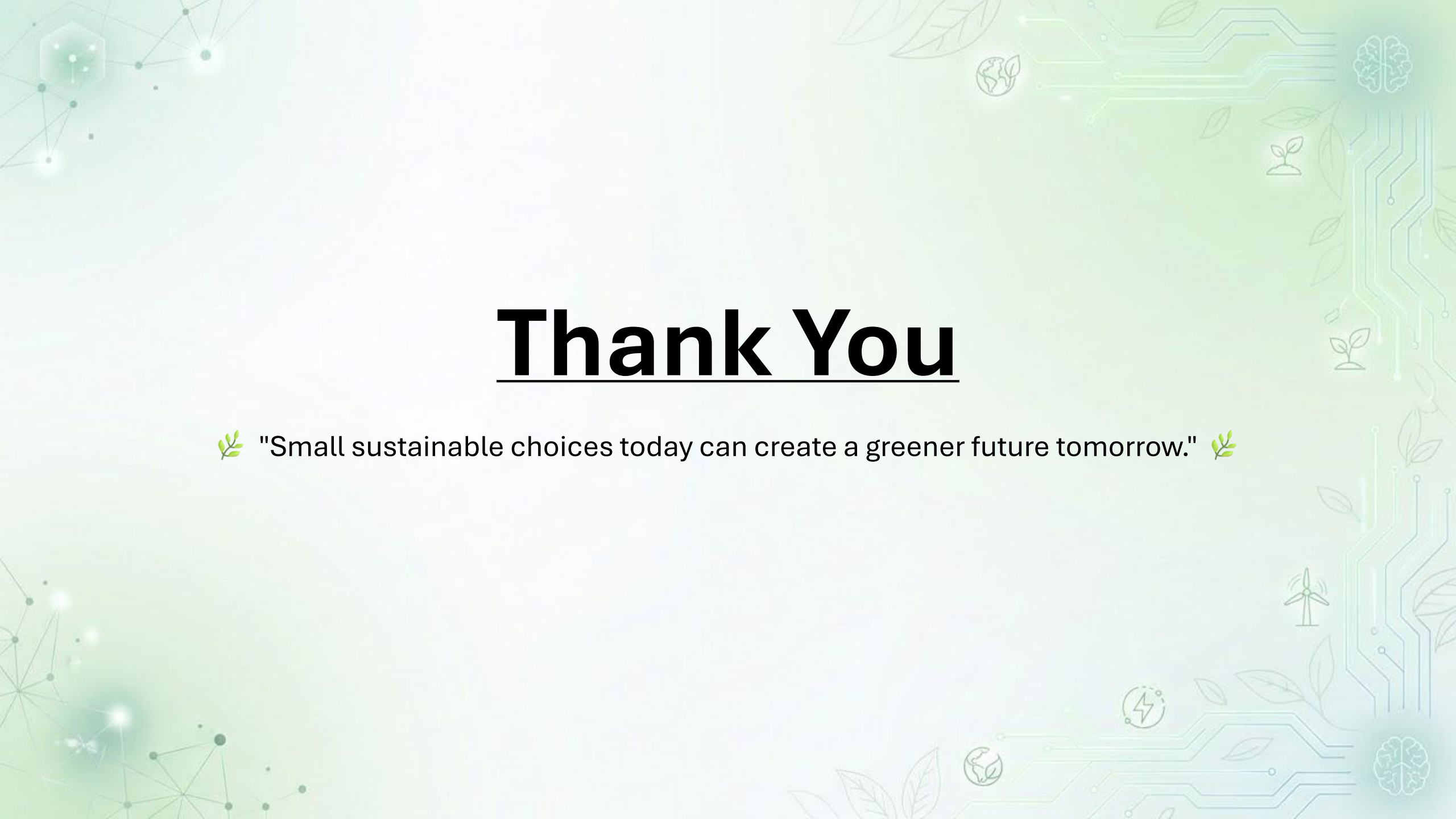
 **Green Future Advisor Agent** demonstrates how AI can support sustainable living by providing personalized environmental guidance.

Key Outcomes:

- Uses AI to make sustainability knowledge accessible
- Helps users understand and adopt eco-friendly practices
- Supports informed decision-making for reducing environmental impact
- Shows the potential of responsible AI in addressing sustainability challenges

Final Message:

AI can be a powerful tool to encourage sustainable actions and create a more environmentally responsible future.



Thank You

🌱 "Small sustainable choices today can create a greener future tomorrow." 🌱